

Rimányi's Weak Chern-Positivity Conjecture for A_5 : An Exact Reduction, Partial Results, and a Handoff for Further Work

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Abstract

This document is a handoff for an agent attempting to prove, or disprove, Rimányi's weak Chern-positivity conjecture for the Morin singularity A_5 . It begins with the setup in `report.tex`, records a new adjacent- transposition reduction and a rigorous proof on the unpaired half of the coefficient cone, and isolates one explicit coefficientwise inequality that would finish the proof. It also records extensive exact evidence, failed certificate searches, and several concrete next directions. Nothing based only on finite expansion is claimed as a proof.

1 Instructions to the next agent

The main target is to prove

$$C(M) \geq 0$$

for every zero-sum multiset M of five integers, with $C(M)$ defined in Section 3. A counterexample is equally decisive and must not be ruled out merely because the computations below are positive.

The most promising current target is the paired inequality

$$A_{i,j,k,l} + A_{j-i,j,k,l} \geq 0 \quad (0 \leq i \leq j), \tag{1}$$

defined precisely below. Proving (1) proves the weak conjecture for A_5 . The complementary range $i > j$ is already proved nonnegative in Proposition 3.

All symbolic computations should use exact rational arithmetic. Before building on a formula in this report, independently verify it from the closed form for F_5 . In particular, do not turn a large truncation into an implicit “proof by computation.”

2 The chamber series

Normalize

$$z_1 = abce, \quad z_2 = bce, \quad z_3 = ce, \quad z_4 = e, \quad z_5 = 1.$$

The chamber expansion from `report.tex` is

$$F_5(a, b, c, e) = \sum_{i,j,k,l \geq 0} A_{i,j,k,l} a^i b^j c^k e^l,$$

where

$$F_5 = \frac{V(1 - 2abce - bce)P}{D}, \tag{2}$$

with

$$\begin{aligned} V &= (1-a)(1-ab)(1-abc)(1-abce)(1-b)(1-bc)(1-bce) \\ &\quad \cdot (1-c)(1-ce)(1-e), \\ P &= 1 - 2abc - bc - ce - bce + 2a^2b^2c^2e + 3ab^2c^2e + 2bc^2e, \end{aligned}$$

and

$$\begin{aligned} D &= (1-2a)(1-2ab)(1-b-ab)(1-2abc)(1-bc-abc)(1-2bc) \\ &\quad \cdot (1-c-abc)(1-2abce)(1-bce-abce)(1-2bce) \\ &\quad \cdot (1-ce-abce)(1-ce-bce)(1-e-abce). \end{aligned}$$

The coefficient $A_{i,j,k,l}$ corresponds to

$$\alpha(i, j, k, l) = (i, j-i, k-j, l-k, -l). \quad (3)$$

The four indices i, j, k, l are the first four partial sums of α .

2.1 The elementary positivity lemma

We repeatedly use the following observation from `report.tex`.

Lemma 1. *If f, g are power series with nonnegative coefficients and zero constant term, then*

$$\frac{1-f}{1-f-g} = 1 + \frac{g}{1-f-g}$$

is coefficientwise nonnegative. Equivalently, $(1-f)/(1-h)$ is coefficientwise nonnegative whenever $h-f$ is coefficientwise nonnegative.

3 The ℓ -free Chern reduction

For a zero-sum multiset M of five integers, define

$$C(M) = \sum_{\substack{\alpha \text{ an ordering of } M \\ \alpha_1 + \dots + \alpha_r \geq 0, 1 \leq r \leq 4}} A_{\beta(\alpha)}, \quad \beta_r(\alpha) = \alpha_1 + \dots + \alpha_r. \quad (4)$$

The reduction proved in `report.tex` says that every Chern-monomial coefficient of the A_5 Thom polynomial, in every relative dimension, is one of the integers $C(M)$. Conversely, every zero-sum M occurs after increasing the relative dimension sufficiently. Hence Rimányi's weak conjecture for A_5 , in all relative dimensions, is equivalent to

$$C(M) \geq 0 \quad \text{for every zero-sum multiset } M \text{ of five integers.}$$

4 The first adjacent transposition

Given a ballot ordering α , interchange α_1 and α_2 . In the $\beta = (i, j, k, l)$ coordinates this is the involution

$$\tau(i, j, k, l) = (j-i, j, k, l). \quad (5)$$

All partial sums after the second are unchanged. The transposed ordering is again ballot exactly when $j-i = \alpha_2 \geq 0$.

Thus the orderings in (4) split into:

- (i) singleton orbits with $i > j$, for which the transposed ordering is not ballot;
- (ii) two-element orbits with $0 \leq i \leq j$;
- (iii) fixed points of τ , which occur when $2i = j$.

It follows that the following two statements would prove $C(M) \geq 0$:

$$A_{i,j,k,l} \geq 0 \quad (i > j), \quad (6)$$

$$A_{i,j,k,l} + A_{j-i,j,k,l} \geq 0 \quad (0 \leq i \leq j). \quad (7)$$

At a fixed point, (7) reads $2A_{i,2i,k,l} \geq 0$, so it also gives the required singleton contribution there.

5 Proved: positivity in the unpaired tail

Remove the only factor that can produce arbitrarily large a -degree without increasing the b -degree:

$$H(a) = \frac{1-a}{1-2a}, \quad J(a, b, c, e) = \frac{1-2a}{1-a} F_5(a, b, c, e). \quad (8)$$

We have

$$H(a) = 1 + \sum_{n \geq 1} 2^{n-1} a^n.$$

Lemma 2. *For every j, k, l , the polynomial*

$$[b^j c^k e^l] J(a, b, c, e)$$

has a -degree at most j .

Proof. After the factor $(1-a)/(1-2a)$ is removed from F_5 , every occurrence of a in a numerator or denominator monomial is accompanied by b . Expanding the remaining denominator factors therefore cannot produce a monomial whose a -degree is larger than its b -degree. $\square$

Proposition 3. *For all $i > j$ and all $k, l \geq 0$,*

$$A_{i,j,k,l} \geq 0.$$

Proof. Write

$$[b^j c^k e^l] J = \sum_{r=0}^j J_r a^r$$

using Lemma 2. If $i > j$, then $i - r \geq 1$ for every $0 \leq r \leq j$, and convolution with H gives

$$A_{i,j,k,l} = \sum_{r=0}^j 2^{i-r-1} J_r = 2^{i-1} [b^j c^k e^l] J(1/2, b, c, e).$$

Direct simplification of (2) yields

$$\begin{aligned} J(1/2, b, c, e) &= \frac{1-e}{1-e-\frac{1}{2}bce} \frac{1-c}{1-c-\frac{1}{2}bc} \frac{1-ce}{1-ce-\frac{1}{2}bce} \\ &\quad \times \frac{1-\frac{1}{2}b}{1-\frac{3}{2}b} \frac{1-\frac{1}{2}bc}{1-\frac{3}{2}bc} \frac{1-\frac{1}{2}bce}{1-\frac{3}{2}bce}. \end{aligned} \quad (9)$$

Each factor in (9) is coefficientwise nonnegative by Lemma 1. Therefore $[b^j c^k e^l] J(1/2, b, c, e) \geq 0$, proving the result. $\square$

Remark 4. *Proposition 3 is an infinite statement. It replaces the previous numerical observation that no negative $A_{i,j,k,l}$ had been seen with $i > j$.*

6 An exact generating function for the paired range

Set

$$S(b, c, e) = J(1/2, b, c, e). \quad (10)$$

For fixed j, k, l , polynomial division at the root $a = 1/2$ gives a polynomial Q defined globally by

$$Q(a, b, c, e) = \frac{(1-a)J(a, b, c, e) - \frac{1}{2}S(b, c, e)}{1-2a}. \quad (11)$$

The apparent pole at $a = 1/2$ is removable coefficient by coefficient, and $\deg_a[b^j c^k e^l]Q \leq j$.

Define

$$\begin{aligned} \mathcal{P}(a, b, c, e) &= Q(a, b, c, e) + Q(a^{-1}, ab, c, e) \\ &\quad + \frac{1}{2} \left(\frac{S(b, c, e) - 2aS(2ab, c, e)}{1-2a} + \frac{2S(2b, c, e) - aS(ab, c, e)}{2-a} \right). \end{aligned} \quad (12)$$

Despite the appearance of a^{-1} , the support bound on Q makes $Q(a^{-1}, ab, c, e)$ an ordinary power series.

Proposition 5. For $0 \leq i \leq j$,

$$[a^i b^j c^k e^l] \mathcal{P} = A_{i,j,k,l} + A_{j-i,j,k,l}.$$

Consequently, coefficientwise nonnegativity of $\mathcal{P}$ proves Rimányi's weak conjecture for A_5 .

Proof. Equation (11) gives, coefficient by coefficient in b, c, e ,

$$F_5 = Q + \frac{S}{2(1-2a)}.$$

The first two terms of (12) reverse the a -degree inside each fixed b -degree. For the remaining terms, if $[b^j c^k e^l]S = S_{j,k,l}$, then

$$\frac{S(b) - 2aS(2ab)}{2(1-2a)}$$

has coefficient $2^{i-1}S_{j,k,l}$ when $0 \leq i \leq j$, and coefficient zero when $i > j$. Similarly,

$$\frac{2S(2b) - aS(ab)}{2(2-a)}$$

has coefficient $2^{j-i-1}S_{j,k,l}$ for $0 \leq i \leq j$, and zero for $i > j$. Combining these identities gives the claim. $\square$

The last line of (12) is coefficientwise nonnegative by (9) and the triangular cancellations described in the proof. The unresolved part is therefore the domination

$$Q(a, b, c, e) + Q(a^{-1}, ab, c, e) \quad \text{against the positive last line of (12).}$$

The first summand alone, the reversed summand alone, and their sum all have negative coefficients.

7 The rational form of $\mathcal{P}$

After reduction, the denominator of $\mathcal{P}$ is a product of the following twenty-one power-series-positive factors:

$$\begin{aligned} &1 - 2b, \quad 1 - 3b, \quad 1 - c - bc, \quad 1 - 2bc, \quad 1 - 3bc, \\ &1 - b - ab, \quad 1 - 2ab, \quad 1 - 3ab, \\ &1 - e - bce, \quad 1 - ce - bce, \quad 1 - 2bce, \quad 1 - 3bce, \\ &1 - c - abc, \quad 1 - bc - abc, \quad 1 - 2abc, \quad 1 - 3abc, \\ &1 - e - abce, \quad 1 - ce - abce, \quad 1 - bce - abce, \quad 1 - 2abce, \quad 1 - 3abce. \end{aligned}$$

The reduced numerator has the visible factors

$$(1 - c)(1 - e)(1 - ce),$$

but the remaining polynomial is large and has mixed signs. Pairing these three factors with denominator factors by Lemma 1 leaves a rational residual whose numerator has 3123 terms, approximately half negative. The residual series itself is not coefficientwise nonnegative: the stripped positive factors are essential to the observed cancellation.

Thus it is not enough to say that all denominator inverses are positive. Any successful rational certificate must retain, or explicitly model, the cancellation supplied by the three simple numerator ratios.

8 A second useful reformulation: prefix sums

Define

$$B(a, b, c, e) = \frac{F_5(a, b, c, e)}{1 - a} = \frac{J(a, b, c, e)}{1 - 2a}. \quad (13)$$

If

$$B = \sum B_{i,j,k,l} a^i b^j c^k e^l,$$

then

$$B_{i,j,k,l} = \sum_{r=0}^i A_{r,j,k,l}, \quad A_{i,j,k,l} = B_{i,j,k,l} - B_{i-1,j,k,l}, \quad (14)$$

where $B_{-1,j,k,l} = 0$.

Equivalently, the scaled generating function

$$W(a, b, c, e) = \frac{J(a/2, b, c, e)}{1 - a} \quad (15)$$

has coefficients

$$[a^i b^j c^k e^l] W = \sum_{r=0}^i 2^{-r} [a^r b^j c^k e^l] J.$$

Large exact computations found every coefficient of B , equivalently of W , nonnegative. This is not yet proved.

In terms of B , the main paired inequality becomes

$$B_{i,j,k,l} + B_{j-i,j,k,l} \geq B_{i-1,j,k,l} + B_{j-i-1,j,k,l}. \quad (16)$$

This suggests looking for a symmetric unimodality, total-positivity, or injection statement for the prefix coefficients $B_{i,j,k,l}$. Merely proving $B \geq 0$ coefficientwise is not by itself sufficient for (16), but it appears to be a useful stronger invariant of the original series.

9 Exact computational evidence

All computations described here used exact integers or rationals.

- The tail statement $A_{i,j,k,l} \geq 0$ for $i > j$ was first observed in large boxes and is now proved by Proposition 3.
- The paired inequality (7) had no failures in every tested box. One large nonrectangular calculation covered 543,507 coefficients, including indices as large as $(10, 45, 35, 35)$.
- Negative Laurent coefficients seen in the tested ranges always had $i \leq j$. Common infinite-looking families include $(i, j) = (1, 1), (2, 2), (3, 3), \dots$, as well as some off-diagonal cases such as $(2, 3)$. Their τ -partners were positive and dominated them.
- Every tested coefficient of $F_5/(1-a)$ was nonnegative.

These observations strongly motivate (7), but none is a substitute for a uniform argument.

10 Approaches tried and what failed

10.1 Other adjacent transpositions

Swapping the second and third increments also passed large finite tests, but its natural kernel is

$$\frac{1-b}{1-b-ab},$$

whose coefficients involve $a(1+a)^{n-1}$; the analogue of the clean evaluation $a = 1/2$ is less transparent.

The third and fourth adjacent swaps fail as pairwise domination strategies. For example, swapping the third and fourth increments pairs

$$(2, 2, 1, 5) \longleftrightarrow (2, 2, 6, 5),$$

and the corresponding coefficients 2 and -3 have negative sum. Swapping the last two increments also fails, including at negative fixed points such as $A_{1,1,2,1} = -1$.

10.2 Greedy positive factorization

The factorization of F_5 in `report.tex` removes ten coefficientwise nonnegative ratios, but its residual series is highly negative. Applying the same idea to $\mathcal{P}$ does not produce a positive residual. Choices obtained by pairing $1-c$, $1-e$, and $1-ce$ with either member of their symmetric denominator pairs were all tested; every resulting residual had negative coefficients.

10.3 A first-order denominator certificate

Let $N/\prod_r(1-f_r)$ be the residual after stripping three elementary positive ratios from $\mathcal{P}$, where every f_r has nonnegative coefficients. A sufficient certificate would be

$$N = P_0 + \sum_r P_r(1-f_r)$$

with all P_0, P_r coefficientwise nonnegative polynomials. The resulting finite linear feasibility problem has approximately 25,000 coefficient constraints and 460,000 nonnegative variables. An exact construction of the system followed by a HiGHS feasibility check showed that this first-order certificate is infeasible.

This does not disprove positivity of $\mathcal{P}$. It says only that a certificate allowing deletion of at most one denominator factor at a time is too restrictive. Products such as

$$P_{rs}(1 - f_r)(1 - f_s)$$

or a positive state-space realization may still work.

10.4 Partial fractions

Partial fractions in a produce poles corresponding to

$$\frac{b}{1-b}, 2b, 3b, \frac{bc}{1-c}, \frac{bc}{1-bc}, 2bc, 3bc,$$

and

$$\frac{bce}{1-e}, \frac{bce}{1-ce}, \frac{bce}{1-bce}, 2bce, 3bce.$$

Individual residues generally have mixed coefficient signs. The pole $3b$ simplifies positively, but the other residues do not, so positivity cannot be proved residue by residue in this decomposition.

10.5 Elementary-symmetric variables

Put

$$x = ab, \quad y = b.$$

Then $a^i b^j = x^i y^{j-i}$, and the involution τ interchanges x and y . The rational series $\mathcal{P}(x/y, y, c, e)$ is symmetric in x, y . Its numerator was rewritten in

$$s = x + y, \quad p = xy.$$

The resulting polynomial still has mixed signs, so positivity is not immediate in the monomial basis in s, p, c, e . A more suitable symmetric basis, or grouping with symmetric denominator pairs, remains possible.

11 Promising next attempts

1. **Prove the paired-series lemma directly.** Work with (12), not with the much larger original Chern sum. Seek a decomposition of $\mathcal{P}$ into a small sum of rational series, each manifestly nonnegative by Lemma 1. Preserve the symmetric x, y structure throughout.
2. **Search for a higher-order finite certificate.** Extend

$$N = P_0 + \sum_r P_r(1 - f_r)$$

by terms $P_{rs}(1 - f_r)(1 - f_s)$, initially only for denominator pairs having the same monomial direction or exchanged by $x \leftrightarrow y$. Use column generation rather than instantiating every possible P_{rs} . If a floating-point solution is found, recover rational coefficients and verify the polynomial identity exactly.

3. **Find a positive state-space realization.** Interpret each inverse $1/(1 - f_r)$ as a branching transition. The failed first-order certificate permits only one transition layer. A finite automaton with several internal states could encode the essential cancellation while keeping all transition matrices nonnegative.
4. **Exploit prefix positivity.** Attempt first to prove

$$\frac{F_5}{1 - a} \geq 0 \quad \text{coefficientwise.}$$

Then investigate whether the coefficient arrays satisfy a stronger monotonicity or total-positivity property implying (16). In the root variables $x = ab, y = b$, the desired statement is a symmetric inequality between adjacent differences of the prefix array.

5. **Use symmetric denominator pairs.** In x, y, c, e , group factors such as

$$(1 - 2x)(1 - 2y), \quad (1 - 3x)(1 - 3y), \quad 1 - x - y,$$

and their c - and ce -multiples before expanding. The basis of complete homogeneous symmetric functions $h_r(x, y)$, rather than monomials in s, p , may match these denominator products better.

6. **Try S_3 -orbit grouping.** Instead of pairing only α_1, α_2 , group all ballot permutations of $\alpha_1, \alpha_2, \alpha_3$. This gives more positive slack and may cancel the chamber kernels associated with the first three roots. Care is needed: only permutations whose intermediate partial sums remain nonnegative belong to the orbit sum.
7. **Classify negative Laurent families only if it becomes finite.** The observed negative coefficients are sparse, but new diagonal families appear as the range grows. If a structural theorem shows that all negatives lie in finitely many parametric families, derive their diagonal generating functions and prove domination by their τ -partners. Do not assume such a finite classification from the current data.
8. **Continue counterexample searches in parallel.** Use the rational form of $\mathcal{P}$, rather than recomputing all of F_5 , to search highly anisotropic ranges and diagonals. A negative coefficient of $\mathcal{P}$ would refute the pairwise strategy, though not necessarily the weak Chern-positivity conjecture itself.

12 Minimal checklist for a claimed proof

A complete proof based on the present reduction should include:

- (a) an exact derivation of (12) from (2);
- (b) a uniform proof that every coefficient of $\mathcal{P}$ is nonnegative, or a different orbit grouping with a corresponding uniform proof;
- (c) explicit handling of fixed points $2i = j$;
- (d) the orbit argument showing that all ballot orderings contributing to each $C(M)$ have been partitioned;
- (e) no appeal to a finite truncation as evidence for an infinite coefficientwise statement.

13 Current status

The following is proved:

$$i > j \implies A_{i,j,k,l} \geq 0.$$

The weak conjecture for A_5 is reduced to the explicit paired inequality

$$A_{i,j,k,l} + A_{j-i,j,k,l} \geq 0 \quad (0 \leq i \leq j),$$

equivalently to coefficientwise nonnegativity of the rational series $\mathcal{P}$ in (12). The inequality has substantial exact computational support, but no proof is currently supplied here.